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AMPN: a semantic resource for Arabic morphological patterns

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Abstract In this paper, we present a pilot Arabic morphological Pattern Net study based on a lexical semantic resource. During this study, a limited number of Arabic Morphological Patterns have been selected in order to analyze the structure and the behavior of the verbs in the Arabic PropBank, which is a semantically annotated corpus of newswire text from the Annahar Journal. Our goal is twofold: (a) to study whether there is a direct relationship between morphological patterns and verbal semantic roles; and, (b) to verify that this direct relationship is a pervasive component of Arabic verb morphology. The approach to building our morphological Patterns database is based on linguistic generalization of the semantic roles of the verbal predicates. The results obtained show promising outcome for a future, more comprehensive study.

We use the Buckwalter Transliteration scheme to represent the Arabic used in this paper http://www.qamus.com.

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1 Introduction

Arabic verbs exhibit both derivational and inflectional morphology. The derivations are known for their template structure. The templates are composed of a combination of roots and patterns, with some degree of idiosyncracy. Roots encode semantic information and they combine with patterns, which mostly encode syntactic information. The resulting templates form lemmas. The roots that form the bases of these lemmas are sets of either three or four consonants, referred to as radicals, which are inserted at specific points within a morphemic pattern. For example, the root d–f–E, "the act of pushing", could be substituted into the numbered radical slots of a pattern [$c^1ac^2c^2ac^3a$] to form a lemma, as shown in the Table 1.

Derived words, or derivations such as *kattaba*, are the lemmas that are further inflected for number, gender, case, mood, aspect, definiteness, etc., to reflect the inflectional morphological system. Typically, verbal pattern morphemes have a set of related 'deverbal' morphemes encoding a variety of syntactic relationships of the term, such as 'location of verb', 'instrument of event'. Thus a single derived form of a verb, such as the combination of the root f–t–H, "to open", and a pattern [$ic^1tac^2ac^3a$], may result in all the deverbals in Fig. 1. The details of how the roots and patterns combine—and how one breaks them down computationally—are valid and difficult questions, ones already thoroughly addressed in work on morphological analysis such as The Challenge of Arabic for NLP/MT Conference (2006), Farghaly and Senellart (2003), and

Table 1 How roots and patterns form lemmas

Lemma	English	Prefix	C1 **d**	Vowel	C2 **f**	Vowel	C3 **E**	Vowel
دَفَعَ dafaEa	To push	_	**d**	a	**f**	a	**E**	a
دَفْعَ dafa ~ Ea	To make push	_	**d**	a	**ff**	a	**E**	a
اِنْدَفَعَ AinodafaEa	To dive into	Aino	**d**	a	**f**	a	**E**	a
دَافَعَ daAfaEa	To defend	_	**d**	a/A	**f**	a	**E**	a
تَدَافَعَ tadaAfaEa	To push one another	ta	**d**	a/A	**f**	a	**E**	a

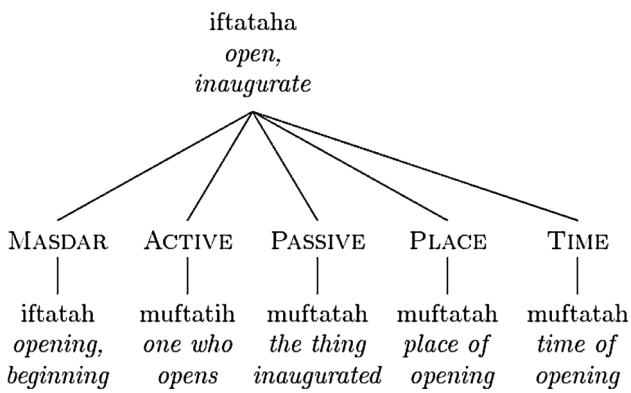

Fig. 1 Deverbals derived from a single combination of root فتح (fth) and pattern (ic^{1}tac^{2}ac^{3}a)

Beesley (2001). This lexical resource therefore will focus entirely upon the semantic and syntactic consequences of these patterns.

In this pilot study, a limited number of Arabic Morphological Patterns have been selected in order to analyze the structure and the behavior of the verbs in the Arabic PropBank (Palmer et al. 2008), which is a semantically annotated corpus. Our goal is to find the relationship between morphological patterns and the verbal semantic roles. In the next section we will present the background on some of the published related research in Arabic Morphology. Section 3 will provide the motivation and the merit behind this pilot study. Section 4 will provide a detailed description of our methodology with the details of the various steps involved in creating this pilot resource.

2 Background

It is worth noting that non-computational classifications of the Arabic morphological pattern system have been investigated in the works of Fehri (1988), Benmamoun (2003), Almutuwwakal (1988) and Watson (2002). Many of the studies in English like Watson (2002) have focused only on a ten core ubiquitously taught 3-letter patterns—patterns for roots composed of three radicals. These studies are incomplete in their coverage since there exist 4-letter roots and, in some rare cases, 5-radical roots, moreover within the 3-letter root pattern template system, there are more than ten 3-letter templates. However, more importantly, while many studies have casually noted various meanings for the patterns, none of these studies did so at the fine level of granularity we are proposing or even in a systematic formal manner, but rather simply assumed a singular conceptual structure for each pattern. Similarly, the Arabic-language work of Fehri (1988) investigates the classes and syntactic alternations of Arabic verbs and adjectives. Yet, it did not distinguish between the inflectional verbal patterns and other derivational morphemes, hence mixing two separate morphological operations (inflection and derivation), and accordingly, masking the regularities that exist latently in the pattern morphology. Almutuwwakal (1988), in contrast, developed a coherent classification of Arabic morphological derivation using the Functional Grammar framework, classifying Arabic verbs according to their argument structure characteristics and derivational patterns. Amutuwakkal's work, however, is entirely theoretical, and therefore, neither grounded in empirical data extracted from a corpus, nor computationally accessible. Yet, it provides a useful set of descriptions against which this pilot study's conclusions could be compared. Accordingly, we are planning to use it as a primary reference point for our proposed investigation. Although there exist a variety of lexical resources for Arabic such as the MUHIT Universal Networking Language (UNL) database Alansary (2014), few provide a systematic investigation of the semantic/syntactic aspects of the morphological system. Resources for morphological analysis, the underlying represented lexica, such as those used in the following systems: Buckwalter (2004), Beesley (2001), Attia (2006), Attia et al. (2011), and Maamouri et al. (2010) allow for the analysis of words into roots and patterns, yet when they provide semantic information via explicit semantic fields, the granularity is at the level of verb lemma. We believe this to be much too coarse grained to exhibit the underlying systematic regularity. The most morphologically detailed of these systems, the system of Attia et al. (2011), treats the morphological pattern as one possible feature for determining a semantic field of a new term. Arabic Wordnet similarly contains information about

semantic relations, but that is also encoded at the lemma level without addressing roots and patterns systematically.

Two more syntactically grounded resources, more relevant to the task at hand, are the Arabic PropBank (APB) (Palmer et al. 2008; Zaghouani et al. 2010), the Quranic Arabic PropBank (QAP) (Zaghouani et al. 2012) and the Arabic VerbNet (Mousser 2011). Arabic PropBank is a corpus of treebanked sentences in which the predicates have been disambiguated and their argument structure labeled. Arabic VerbNet, based upon the English VerbNet project (Kipper et al. 2008), is a mapping of Levin's verbs syntactic alternations and classification into Arabic (Levin, 1993), with alternations added whenever necessary. This provides a rough verbal classification of syntactic alternations, with the same semantic and syntactic representations of these classes as utilized in English VerbNet. One complexity of Mousser's Arabic VerbNet, however, arises from the richness and power of the morphological system itself. Syntactic alternations in English—such as the difference between "John dried the clothes" and "the clothes dried"—may be expressed in Arabic using different, yet very similar verb lemmas—in this instance, našafa and naššafa respectively (example from Mousser 2011, p. 1). As such, Mousser introduces "sibling classes", essentially treating certain morphological variants as part of the same syntactic alternation (Mousser 2011). Hence, it is, on the one hand, fine grained in representing shared syntactic and semantic information across verb lemmas, yet coarse grained in grouping some of these lemma templates. Crucially however, we believe, that this exemplifies the complexity of dealing with such a morphologically-rich language without having a core resource systematically modeling the intricacies of the underlying morphology.

These lexical resources have, however, given Arabic NLP enough access to the Arabic morphological system albeit in perhaps a simplified manner—to allow basic forays into its use. Explicitly modeling aspects of the morphological system in NLP has led to improvements in performance even when using simplistic representations of the root pattern system or explicit encodings of morphological features as evidenced in tasks such as Information Retrieval (Al-Ameed et al. 2005) and Arabic Semantic Role Labeling (Diab et al. 2007; Diab and Moschitti 2007; Diab et al. 2008; Palmer et al. 2008). Such NLP systems would presumably see even better improvements if they had access to richer lexical resources for Arabic morphology, especially if there existed a VerbNet style database explicitly encoding syntactic regularities at the pattern sense level of granularity proposed here. Counter intuitively, several systems that attempted to explicitly represent a 'pattern' feature, given the current resources, failed to show relative improvement in performance. For example, the automated verb classification system described in Snider and Diab (2006) and the dependency parsing of (Marton et al. 2010) found no improvements in the results despite the inclusion of an explicit 'pattern' feature. This may be because each disambiguated pattern packs a vast range of syntactic alternations that are not correctly identified for such systems. The level of granularity for the pattern encoding was too coarse grained. The results of these systems findings are not discouraging, but rather evidence that the morphological system of Arabic is too complicated to approximate without semantically rich lexical resources. A system such as Snider and Diab (2006) would have benefited tremendously from a computationally accessible pattern sense inventory with associated consistent sets of syntactic alternations.

3 Motivation

The Arabic Morphological Pattern Net (AMPN) aims at representing both the complex, derivational structure for the morphological root patterns of Arabic, while also dealing with the multiple senses within each of these patterns in turn. These morphological roots (which carry the semantic content) combined with the patterns that regulate the syntactic clausal behavior of the lexicon) added together provide the 'lemma' of the verb. Each verb lemma, in turn, would be marked for the particular sense of its derivational pattern. In this respect, it is presumed that a system addressing these issues would base its conclusions upon actual verbs in context. It is further noted that the regularities within each sense noted above all concern issues of semantic form, selectional restriction, and argument structure. A resource representing these senses and their derivations would thus need to represent these regularities in a consistent manner, such as the one used in systems like VerbNet (Kipper et al. 2008).

Such a resource could be invaluable for Arabic NLP and could help address undisambiguated pattern feature that has been useful to date in Arabic NLP tasks. This, in turn, would enhance the robustness of Arabic NLP when it comes to the analysis of the Arabic morphological system that has always been a major challenge for NLP (Soudi et al. 2007).

4 Methodology

The following assumptions underline our choice of representation:

(1) Morphological templates, though non-contiguous, are handled like morphemes, and given lexical entries as such; (2) these morphemes are polysemous, and their different senses will display different syntactic alternations;

(3) each sense will be represented using a list of which verb roots which combine with that sense.

The suggested system of analysis depends on lexical entries of verb lemmas that consist of a combination of roots and morpheme patterns rather than individual words. These lexical entries are analyzed along a number of dimensions, to capture the various semantic, syntactic, and semantic layers of the pattern. Accordingly, we started from the APB (Palmer et al. 2008; Zaghouani et al. 2010).

The APB consists of two major portions: the lexicon resource of Frame Files and the annotated corpus. To show the components of the APB, we illustrate in Fig. 2 an example of a Frameset and in Fig. 3 and annotation example. The frameset in APB contains the English definition of the verb, the argument structure for the actual frameset (e.g. Arg0 for the subject and Arg1 for the object etc.) and at least a parsed Arabic example in Treebank format used as an illustration. Figure 3 illustrate a sentences tagged by the annotators for the given frameset, the annotation will be based on the given argument structure.

To further investigate our assumptions, we selected 2100 APB verb frames and looked at their application to ten specific morphological Patterns. The selected patterns represent three levels of the assumed derivational hierarchy as follows: (1) C1aC2C3aC4; (2) C1aC2C2aC3, C1aaC2aC3, CaC1C2aC3, C1taC2aC3, nC1aC2aC3 and stC1C2aC3; and (3) taC1aC2C2aC3, taC1aaC2aC3 and taC1aC2C3aC4.

Firstly, we annotated the APB verb file (2100), added a Morphological Pattern and root to each verb (verb lemmas), and made a subfile for each different Morphological Pattern (MP) that can form a verb. For each MP, we grouped its possible frames of analysis using many factors that include: (1) the syntactic alternations of the verb; (2) number of arguments; (3) nature of arguments; (4) the semantic features of the arguments; (5) the semantic roles, and semantic classification; as well as (6) aspectual types. It is worthy of note that the verb can exist within more than one frame of analysis. Accordingly, we adopted some verb syntactic characteristics to classify the Arabic verbs based on their syntactic alternations. This, in turn, allowed us to grasp the different senses that each morphological pattern may have. An example of how this is organized can be seen in Fig. 4, which shows all of the lemmas formed by a single root, **f–t–H**.

In depending on the APB, we employed two major features: Firstly, the argument structure of the

Predicate: **{isotamaE** استمع
Roleset id: f1, to listen
Arg0: entity listening
Arg1: thing listened

Fig. 2 The frameset of the verb {isotamaE (Zaghouani et al. 2010)

Rel: {isotamaE, استمع
Arg0: -NONE- *
Gloss: He
Arg1: الى مطالبهم
Gloss: to their demands
Example: استمع الى مطالبهم

Fig. 3 An example annotation for a sentence containing the verb {isotamaE (Zaghouani et al. 2010)

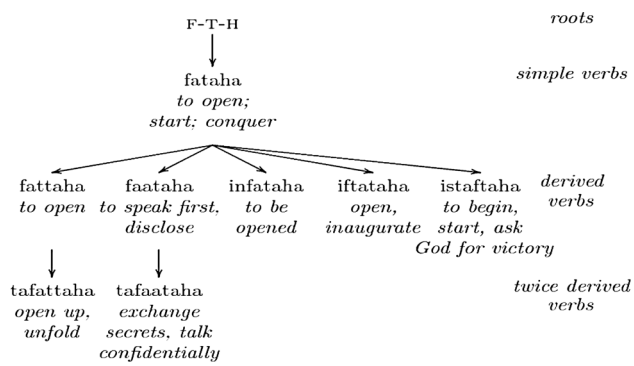

Fig. 4 The derivation of all verb lemmas for the root ف-ت-ح f–t–H

morphological pattern; that is, distinguishing the number of arguments for each verb (ARG0 or ARG1). In addition, we employed the lexical representation of the APB to explore the nature of the verb arguments by exploring the semantic roles as well as the semantic features of arguments in terms of 'rationality' (Rational/Irrational/Both); and number (Singular/Plural/Both) with the exception of the simple trilateral verb.

4.1 Annotation methods

The details of how we implemented our proposal involve three different kinds of annotation that have some interdependencies: (a) clustering predicate PropBank senses into specific senses for specific morphological patterns; (b) developing detailed lexical entries for each sense in the created clusters (it is worth noting that each of these clusters is linked to numerous verbal and deverbal entries via PropBank); (c) linking these senses to each other, so as to model semantic derivation rules, shared syntactic alternations for the different patterns to discover possible derivational gaps.

4.2 Clustering into different senses

The first task of clustering predicate PropBank senses into specific senses for specific morphological patterns begins with a quick annotation of all predicate instances within the

Arabic PropBank corpus for their morphological patterns. We performed this task manually for ten morphological patterns, but it is a relatively simple task, which may be preprocessed later on using some automatic taggers using morphological disambiguators such as MADAMIRA (Pasha et al. 2014), MADA (Habash and Rambow 2005), or AMIRA (Diab 2009), followed by a stage of manual correction.

4.3 Lexical entries for each sense

The second group of tasks is the creation of detailed lexical entries for these senses. We performed this task manually for our ten verbal patterns. This has some similarity to verb classification systems such as VerbNet (Kipper et al. 2008), and Arabic VerbNet (Mousser 2011). Therefore a roughly similar formalism is used for the main elements of these entries: description of syntactic alternations, the expression of thematic roles, and the use of semantic notation to express the semantics of a pattern.

The annotation of syntactic alternations is based upon sentences in the Arabic PropBank corpus, which are examined to reveal the syntactic features of all the verbs in a given sense. As each sense of a pattern is presumed to be syntactically coherent, each set of alternations should be different than those of the other senses. The vocabulary of roles used in this annotation is the same set of 30 thematic roles used in the current VerbNet project (Kipper et al. 2008), and the same notation of syntactic restrictions (currently 41).

The syntactic alternations follow VerbNet conventions with the exception of the Arabic 'subject', which can have a number of positions, none of which seem to be determined by verbal syntax, while the order of Arabic is canonically VS(O), one finds a great deal of 'topicalized' SV(O), pro-drop V(O), and an unusual number of SSV(O) constructions equitable to the English "John he went to the store". While there is no consensus about how to predict these forms (cf. Abdul-Roaf 1997), no claims seem to link these alternations to the verbs involved, and we therefore believe that this four-fold increase in syntactic alternations can be simplified and ignored for this resource. Thus subject position is always treated as being in the canonical VS (O) position after the verb, which parallels the treatment in the Arabic Treebank, by treating SVO position as tropicalized. The resultant form of syntax, for something such as the reflexive form of [ic1tac2ac3a], shows:

(1) Verb Actor (**Agent** + **Patient**) <+plural>

(2) Verb Actor (**Agent** + **Patient**) <+plural > {preposition} (Topicalized)

We also propose annotating for grammatical aspect as implied by the pattern. Each pattern conveys not simply valence information, but also specific grammatical aspect, with many senses of the simple morphological patterns roughly corresponding to Vendler *Aktionsart* classes ([C1aC2uC3-u] considered generally to be stative, [C1aC2iC3-a] considered to be accomplishments, etc.) and more complicated patterns involving other aspects, so that [taC1aaC2aC3a] may involve gradual repeated events (such as tat_abaa, to follow one another consecutively). The relationship between the annotation of the grammatical aspect of these senses, and the semantic representation of the same senses is kept separate (despite the theoretical capacity of a semantic representation such as that of VerbNet to handle aspectual complexity) so as to avoid complexities involving how grammatical aspect is inherited. Finally, using a conceptual semantic structure similar to VerbNet, models for the semantics of a given pattern are also presented. However, unlike verbal classifications which presume some semantic coherence is noted and that the roots act like a variable within these structures. As such, these formulate semantic structures with actual spaces left for the root. One may compare the VerbNet patterns for the *eat* and *feed* classes below, with some Arabic words for *to eat* and *to feed*. We see that by characterizing the patterns for generative derivations, the meaning of large amounts of terms may be generated.

(3) Pattern **C1aC2aC3a.1:**

Root (during (E), Agent, Patient) + Root akl: take_in = akala:
take_in (during (E), Agent, Patient)
+ root **tQm**: **take_in** = **taQama**: **take_in** (during (E), Agent, Patient)
Comparable to English VerbNet: **eat-39.1:** take_in (during (E), Agent, Patient)

(4) Pattern **aC1aC2aC3a:**

Root (during (E), Recipient, Patient) cause (Agent, E) + Root akl: take_in = aakala:
take_in (during (E), Agent, Patient) cause (Agent, E)
+ root **tQm**: **take_in** = **ataQama**: **take_in** (during (E), Agent, Patient) cause (Agent, E)
Comparable to English VerbNet: **feeding-39.7**: take_in (during (E), Recipient, Patient) cause(Agent, E)

As such, one may see that while this pattern lexicon does not form the same sort of semantically coherent verb classifications seen in a VerbNet-style approach, one could easily collect these syntactic patterns with semantically

similar roots to form semantically coherent verb classes. How this would compare with other automated verb classifications, such as (Snider and Diab, 2006) remains to be seen.

Finally, these entries would also be developed for the "deverbal" morphemes, such as the verbal nouns, active participles, etc., for each sense. These deverbal morphemes would mostly be automatically derivable from their originating patterns, and would otherwise be done by hand in a fashion similar to the above methods.

4.4 Annotating patterns of derivation

Finally, a third set of tasks would involve marking the derivational linkages between these various entries, both semantic and syntactic. This would involve going through each lexical entry described in Sect. 4.3, gathering data concerning its presumed derivations (subtle shifts in semantics being a core marker of this phenomenon), and marking direct linkages to other senses in other patterns, based upon the data at hand and existing works (Al-Najem 2007). For example, while it is well known that "pattern V", taCaCCaCa, is often derived from "pattern II", CaCCaCa, it really only derives from a subset of senses within pattern II. This would contribute additional value to the larger task of identifying the sense of a given morpheme pattern, as it allows one to use the lexicon in order to obtain more accurate disambiguation of novel verbal instances.

This linkage would be applied to the deverbal forms as well. This would provide mappings for how deverbal forms develop, while also indicating which senses are more productive for various deverbal forms. This would not only aid in the task of disambiguating new words, but could also aid in simpler tasks of morphological analysis, adding a more nuanced statistical measure of probability to derivation rules which are currently represented only as possible or impossible.

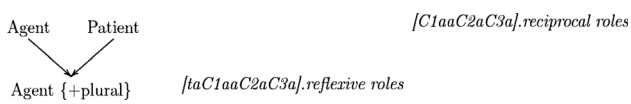

Fig. 5 Mapping the derivation of the thematic roles

Table 2 Example verb-senses

Verb	Transliteration	English sense
اختفى	AixotafaY	To disappear (from s.t.)
اغتسل	Aigotasal	To bathe o.s
ابتعد	AibotaEad	To refrain (from s.t.)
امتنع	AimotanaE	To content o.s (with s.t.)
اكتفى	AikotafaY	To exhaust o.s (with s.t.)
اجتهد	Aijotahad	To retire/withdraw o.s (from s.t.)

These linkages would not simply mark one term as being derived from another, but would also map semantic role labels onto each other. As such, marking the derivation which starts with the reciprocal sense of [C1aaC2aC3a] (having transitive syntax for phrases like 'John quarreled with Mary') and results in the reflexive [taC1aaC2aC3a] pattern (with intransitive meanings like 'John and Mary quarreled'), this sort of coding would occur in the reflexive

Table 3 Summary results of the pilot study of the AMPs

Level	MP (# of frames in APB)	Frame	Verb freq.
2	اسْتَفْعَل AsotafoEal	F_1: Causative Verbs	98
	staC1C2aC3 (179)	F_2: Ergative Verbs	96
		F_3: Inchoative Verbs	52
		F_4: Reflexive Verbs	23
2	اِنْفَعَل AinofaEal	F_1: Causative Verbs	6
	nC1aC2C3 (111)	F_2: Ergative Verbs	33
		F_3: Inchoative Verbs	21
		F_4: Reflexive Verbs	61
2	اِفْتَعَل AifotaEal	F_1: Causative Verbs	131
	ftaC1aC2C3 (342)	F_2: Ergative Verbs	40
		F_3: Inchoative Verbs	42
		F_4: Reciprocal Verbs	16
		F_5: Reflexive Verbs	113
2	أَفْعَل OafoEal	F_1: Causative Verbs	265
	ʔaC1C2aC3 (289)	F_2: Inchoative Verbs	15
		F_3: Reflexive Verbs	9
2	فَاعَل faAEal	F_1: Causative Verbs	59
	C1aaC2aC3 (146)	F_2: Inchoative Verbs	4
		F_4: Reciprocal Verbs	82
3	تَفَاعَل tafaAEal	F_1: Causative Verbs	18
	taC1aaC2aC3 (125)	F_2: Ergative Verbs	5
		F_3: Inchoative Verbs	8
		F_4: Reciprocal Verbs	82
		F_5: Reflexive Verbs	10
2	فَعْل faE ∼ al	F_1: Causative Verbs	352
	C1aC2C2aC3 (357)	F_4: Reciprocal Verbs	2
		F_4: Reflexive Verbs	3
3	تَفَعْل tafaE ∼ al	F_1: Causative Verbs	26
	taC1aC2C2aC3 (225)	F_2: Ergative Verbs	64
		F_3: Inchoative Verbs	68
		F_4: Reciprocal Verbs	33
		F_5: Reflexive Verbs	64
1	فَعْلَل faEolal	F_1: Causative Verbs	15
	C1aC2C3aC4 (22)	F_2: Inchoative Verbs	1
		F_3: Reciprocal Verbs	1
		F_4: Reflexive Verbs	5
2	تَفَعْلَل tafaEolal	F_2: Ergative Verbs	3
	taC1aC2C3aC4 (8)	F_4: Reflexive Verbs	5

Table 4 Example of the frames of the verb افتعل **AftEl**

Frame	Type	Examples	Argument structure	Modifiers	Event type
Frame_1	Causative Verbs	**اِغْتَنَم {igotanam**	Arg0_Agent_RS Arg1_Patient/Theme_BB	Preposition (من/عن/إلى)	Accomplishment/ Achievement
Frame_2	Ergative Verbs	**اِمْتَلأَ {imotalaO** **اِحْتَرَق/ {iHotaraq**	Arg1_Theme _BB Arg2_substance/location_IB	Preposition (بـ)	State/Activities
Frame_3	Inchoative Verbs	**اِعْتَلّ{iEotal ~ /اِخْتَمَر{ixotamar**	Arg1_Experincer_BB Arg2_IB (Beneficiary/Manner)	Preposition (من/بـ)	State
Frame_4	Reflexive Verbs	**اِعْتَزَل {iEotazal**	Arg0_Agent_RB Arg1_Location_BB	Preposition (في/ب)	Achievement/ Activity
Frame_5	Reciprocal Verbs	**التقى AltqY/** **اجتمع AjtmE**	Arg0_Agent/Actor_BP Arg1_Topic_IB	Preposition (بـ)	Activity/ Achievement

[taC1aaC2aC3a] sense as shown in the mapping of the thematic roles in Fig. 5.

4.5 An example output

Thus one would end up with a full lexical entry such as the following:

iC1otaC2aC3a-1
Prose description:

Reflexive

Roles:

Arg0: Agent <+int_control > = **Theme**
Arg1: location

Syntax:

V Agent = Theme
V Agent = Theme {preposition} (**Location**)

Semantics:

ROOT (during (E), Agent, Theme)

Derived from senses:

[C1aC2aC3a-yaC1aC2iC3]
[C1aC2aC3a-yaC1aC2uC3]
[C1aC2aC3a-yaC1aC2aC3]

Table 2 illustrates pattern verb senses. These patterns look longer because a set of the simple verb paradigms are the same in the perfective form and require marking of the imperfective to distinguish them.

Finally, the Table 3, shows the results of our study of ten morphological patterns through the annotation of the proposed AMPN verb frames for all the verbs to which the examined patterns apply, and included in the APB. The first column represents the three levels of the assumed derivational hierarchy, the second column represents the morphological patterns and number of verb frames that appeared in APB files, the third column represents the verb pattern frames classified according to their syntactic alternations, and the fourth column includes the number of APB verb frames that follow each verb pattern frame. Table 4 illustrate one example of an entry for the verb افتعل AftEl.

5 Conclusion and future work

The present paper introduced a pilot study for a Lexical Semantic Resource for Arabic Morphological Patterns. The goal of this project was to represent syntactic valency and alternation of Arabic morphological patterns. The current pilot work was built on the Arabic PropBank for the Arabic Verb Lexicon (2100 Verb Frames). We annotated and classified the APB verbs based on their argument structure and aspectual features. The result of our analysis represents the possible frames for each pattern, which could be considered an infrastructure for an Arabic Lexicon. Our future work is to build a full-sized resource for Arabic Morphological Patterns that covers verbs and the derivational lexicon, to build a comprehensive lexical semantic resource for AMPN, and to map it to several lexical resources, such as the Arabic Propbank, the Arabic VerbNet, and the Arabic WordNet.

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